Project 3: Ion Drive Rocket Analysis

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The purpose of this project is to

Nomenclature

T	=	temperature $[K]$
P	=	pressure $[Pa]$
ρ	=	density $\left[kg/m^3\right]$
T_t	=	total temperature $[K]$
P_t	=	total pressure $[Pa]$
u	=	velocity $[m/s]$
M	=	mach number
A	=	area $\left[m^2\right]$
F	=	thrust $[N]$
R	=	gas constant $[J/kgK]$
C_p	=	constant pressure specific heat $[J/kgK]$
γ	=	specific heat ratio
$\dot{m}$	=	mass flow rate $[kg/s]$
I_{sp}	=	specific impulse $[s]$
g_0	=	standard gravitational acceleration $\left[9.81m/s^2\right]$

I. Introduction

The purpose of this project temperature and pressure information is shown in table **??**. This data was used when running the solver at each altitude.

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II. Methodology

A. Rocket Analysis

The following equation can be used to calculate the propellant mass required to perform a maneuver.

$$\frac{\Delta v}{C} = \ln\left(\frac{m_0}{m_0 - m_p}\right) - \int_{t_0}^{t_f} \frac{D}{Cm} dt - \int_{t_0}^{t_f} \frac{g\cos\theta}{C} dt \tag{1}$$

Because the drag term and the gravity can be ignored, this equation can be simplified to the following.

$$m_p = m_0\left(1 - e^{\frac{-\Delta v}{C}}\right) \tag{2}$$

C in this equation is the effective exhaust velocity which can be found by using the following.

$$I_{sp} = \frac{C}{g_0} \tag{3}$$

B. Ion Drive Analysis

$$I_{sp} = \frac{1}{g_0}\sqrt{2\frac{q}{m}(\phi_A - \phi_C)} \rightarrow (\phi_A - \phi_C) = V \tag{4}$$

$$e \tag{5}$$

C. Brayton Cycle Analysis

$$\pi_t = \frac{1}{\pi_c} \tag{6}$$

$$\frac{T_4}{T_3} = 1 - \eta_t\left(1 - \pi_t^{\frac{\gamma-1}{\gamma}}\right) \tag{7}$$

$$\frac{T_2}{T_a} = 1 + \frac{1}{\eta_c}\left(\pi_c^{\frac{\gamma-1}{\gamma}} - 1\right) \tag{8}$$

$$\eta = 1 - \frac{T_4 - T_a}{T_3 - T_2} \tag{9}$$

$$\frac{A_{rad}}{W_{out}} = \frac{\left(\frac{1}{\eta} - 1\right)}{3(T_4 - T_a)\sigma\epsilon}\left(\frac{1}{T_a^3} - \frac{1}{T_4^3}\right) \tag{10}$$

III. Results

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IV. Conclusion

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V. Appendix A